SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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Multithreading Practice - CO3

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QUESTIONS

Q1
Multiple Threads Using a Single Class
10 marks

Q2
Multiple thread classes
20 marks

Q3
Thread Synchronization
10 marks

3/3 answered

Q3 Thread Synchronization

10 marks

opic: synchronized, shared resource, race condition

Question:
Write a Java program to demonstrate thread synchronization using a shared Counter object.

Create a class Counter containing an integer variable count, initially set to 0.

Create two threads, ThreadA and ThreadB. Both threads should increment the same count variable 1000 times.

Implement the increment() method using the synchronized keyword so that only one thread can modify the counter at a time.

After both threads complete their execution, display the final value of count.

Expected Output:

ThreadA completed
ThreadB completed
Final Count = 2000

Requirements:

Use a shared Counter object.
Both threads must access the same Counter object.
Use synchronized for the increment operation.

Use join() to ensure both threads finish before displaying the final count.

Your Code

```
class Main {
    public static void main(String[] args)

        counter co = new counter();

        ThreadA T1 = new ThreadA(co);
        ThreadB T2 = new ThreadB(co);

        T1.start();
        T2.start();

        T1.join();
        T2.join();

        System.out.println("Final Count =
    }
}


class counter {

    int count = 0;

    public synchronized void counterIncrem
        count++;
    }
}


class ThreadA extends Thread {

    counter c;

    ThreadA(counter co) {
        c = co;
    }

    public void run() {

        for (int i = 0; i < 1000; i++) {
            c.counterIncrement();
        }
```

```
        System.out.println("ThreadA comple
    }
}


class ThreadB extends Thread {

    counter c;

    ThreadB(counter co) {
        c = co;
    }

    public void run() {

        for (int i = 0; i < 1000; i++) {
            c.counterIncrement();
        }

        System.out.println("ThreadB comple
    }
}
```

Input

```
stdin input
```

Output

```
ThreadA completed
ThreadB completed
Final Count = 2000
```

Visible Test Cases

Test Case 1 ✔

Test Case 2 ✔

Test Case 3 ✔

10.00 / 10 marks

Submitted